IAEA Coordinated Research Project on Development of Harmonized QA/QC Procedures for Maintenance and Repair of Nuclear Instruments

Test Procedure for Single Channel Analyzers

PROCEDURE Nº MRNI-512 REV. D0

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1.- OBJECTIVE AND SCOPE

1.1.- Objective

The objective of this procedure is to describe the test procedures for the verification of performance and measurement of the characteristics of Single Channel Analyzers, SCA, used in pulse height analysis, PHA, and counting in radiation detection chains of signals obtained from radiation detectors.

1.2.- Scope

These procedures are applicable to SCA used in Nuclear Instrumentation.

2.- NOTATION AND DEFINITIONS

2.1.- Notation

NIM	Nuclear Instrumentation Modules.
LLD	Lower level discriminator
ULD	Upper level discriminator
DNL	Differential non linearity
INL	Integral non linearity
PHA	Pulse height analysis
E	Main voltage discrimination level related with the amplitude of a signal
ΔE	Window, voltage discrimination gap for signals to be considered in the analysis.
INT	Integral discriminator, integral mode
DIFF	Differential mode, SCA mode

2.2.- Definitions

2.2.1. Detector
A device that converts the energy of a photon or incident particle in an electric pulse.

2.2.2. Differential non linearity, DNL
Value that determines the maximum variation in channel width in a SCA.

2.2.3. Integral non linearity, INL
Value that determines the maximum deviation from a straight line in the relationship between discrimination level and applied voltage in a SCA.

2.2.4. Single Channel Analizer, SCA.
Nuclear instrument that provides a digital pulse every time the analog input pulse fulfills the established requirements of amplitude.

3. INTRODUCTION

The measurement of the height of the pulses produced by a radiation detector provides the information about the energy of the incident radiation, because in most of the cases, the pulse height is proportional to the energy delivered in the detector.

When a selection of defined pulses according with its amplitude is required, the SCA can make this selection. Every time the analog input pulse fulfills the established requirements of amplitude, the SCA provides a digital pulse to be counted.
The SCA can be operated as integral discriminator, INT, see Fig 1, only the pulses with amplitude above the lower level discriminator, LLD or E, will produce an output pulse. LLD can be adjusted as desired.

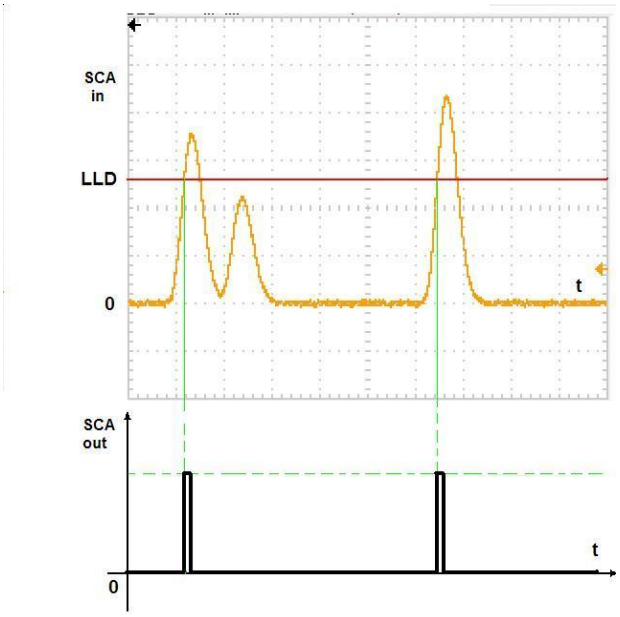


Fig. 1. Integral mode of operation of a SCA.

The SCA can also be operated in differential mode, DIFF, see Fig 2, only the pulses with amplitude between LLD and the upper level discriminator, ULD, (this gap is also called window, ΔE) will produce an output pulse. LLD and ULD can be adjusted as desired.

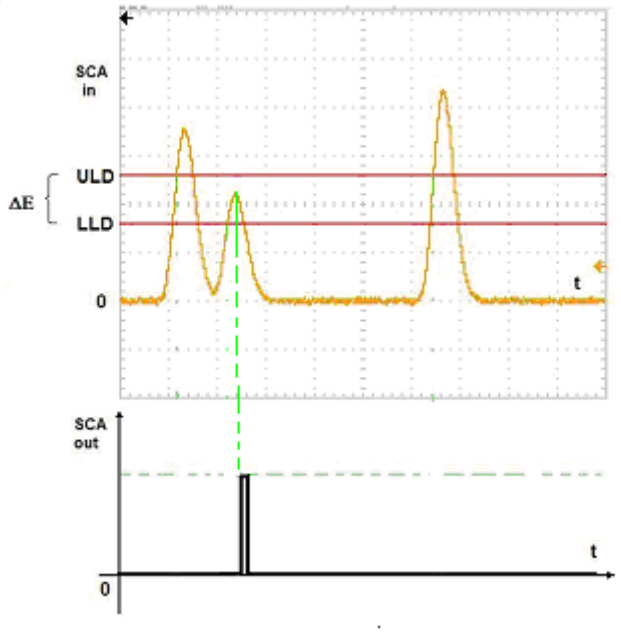


Fig. 2. Differential mode of operation of a SCA.

4.- DEVELOPMENT

The sequence of steps to verify the electrical characteristics of SCA´s is described in the next paragraphs. A flux diagram of the process is shown in Annex I.

4.1- Generalities

Whenever possible, refer to the test conditions recommended by the manufacturer in the sheet of specifications, a technical report about the verification of the electrical characteristics of the SCA must be elaborated, including the circuit diagram, environmental conditions, testing set-up, and details of the test instruments.

4.1.1.- Verify that the SCA is installed and connected according to the instructions of the manufacturer.

4.2.- Test Instruments

All the instruments employed in the tests must be calibrated and with a valid calibration certificate.

4.2.1 Pulse generator.

Produces voltage tail pulses with a precise relative amplitude that is selected with a calibrated dial and attenuation steps.

4.2.2 Spectroscopy amplifier.

Amplifies tail pulses and includes a semi-gaussian shaping in the pulse, use the unipolar output. A good long term stability is desired, i.e. better than 0.01 %/ºC.

4.2.3 Oscilloscope

Use an analog or digital oscilloscope.

4.2.4 Counter or Scaler

The number of counts per unit of time is measured with a counter or scaler for nuclear pulses, it generally includes a voltage discriminator to block low amplitude noise pulses. The input pulses to the counter or scaler must be positive.

4.3- Test Conditions

Generally the SCA needs to reach its stable temperature, in order to have its best performance, a warming period of one hour is enough.

4.3.1.- Temperature

A stable temperature is recommended to perform the tests, between 20º C and 25ºC.

4.4.- Measurements

4.4.1. Characteristics of the output pulse.

Assemble the set-up of Fig. 3, the shaping time in the amplifier is fixed to 6 μs to reduce any influence of a fast rising time of the pulse, use its unipolar output, the SCA is used in the INT mode, put the LLD at the minimum, apply pulses from the generator to the input of the SCA through the amplifier, see the pulses in the output of the SCA with the oscilloscope, measure its characteristics.

4.4.2. Measurement of the integral non linearity of the LLD in INT mode.

Using the set-up of Fig. 3, follow the next steps:

- Adjust the amplitude of the generator to get the minimum signal at the output of the amplifier.
- Put the LLD dial of the SCA in E = 0.5, increase carefully the amplitude of the generator until a SCA output signal just can be seen in the oscilloscope, measure the amplitude, Ve, of the analog pulse at the input of the SCA. Record this value in the Table 1 of the Test Report.
- Increase the value of E according to the Table 1 and repeat the same procedure.
- Make a plot of the E settings as a function of input voltages.
- Analyze all the values of Table 1 with a reliable software, for example ORIGIN, to get the linear fit and the deviation of voltage values with respect to the ideal ones obtained from the linear fit. See Fig. 4.

$$Y = m x + b \qquad (1)$$

Where:

Y = values of the input voltage x = LLD values
m = slope of the line: voltage / LLD values b = voltage for zero LLD

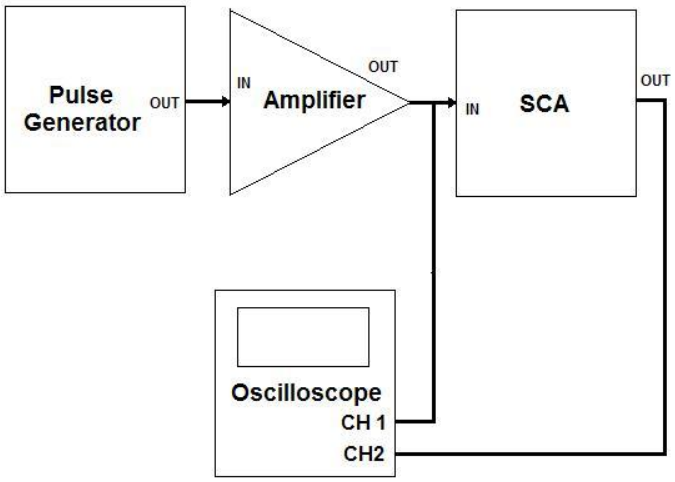


Fig. 3. Set-up for the measurement of the linearity of the LLD dial.

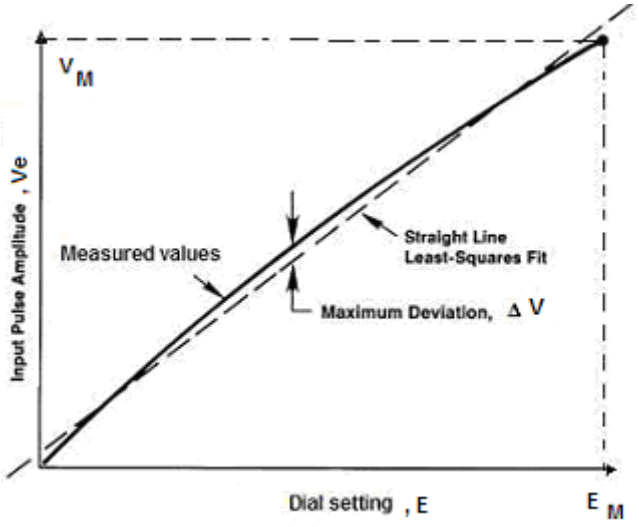


Fig. 4. Graphical representation of the INL. Adapted from: www.ortec-online.com/electronics/adc/intro8.htm

- For every dial setting Ei we have the real value Ve of the input voltage and the ideal value Vi. The maximum deviation, ΔV_{max}, obtained along the total range V_M, defines the value of the INL in per cent:

$$INL = \frac{(V_e - V_i)\max}{V_M} \times 100 \qquad (2)$$

Where:
Ve- input voltage obtained in the real measurement.
Vi – ideal voltage obtained from the linear fit.
V_M – Maximum input voltage corresponding to the maximum Dial setting.
$\Delta V_{max} = (V_e - V_i) \max$

4.4.3. Measurement of the precision of the window in a SCA.

The window is defined when the SCA is operated in DIFF mode, in this test the precision of the window along all the range of E values will be evaluated. Using the set-up of Fig. 3, follow the next steps:

- Adjust the amplitude of the generator to get the minimum signal at the output of the amplifier.
- Fix the window value to 0.5.
- Put the LLD dial in E = 1.0, increase carefully the amplitude of the generator until a SCA output signal just can be seen in the oscilloscope, this is the amplitude V_L, of the analog pulse at the input of the SCA, measure and record this value in the Table 2 of the Test Report.
- Increase carefully the amplitude of the generator until the SCA output signal just disappears fro the oscilloscope screen, this is the amplitude V_U, of the analog pulse at the input of the SCA. Measure and record this value in the Table 2 of the Test Report.
- Repeat the process for the different values of E.
- Analyze all the values of Table 2 to get the maximum deviation of ΔV with respect to the calculated mean value, Vm.
- The maximum deviation, ΔV_{max} obtained, defines the value of the precision, P, of the window in per cent:

$$P = \frac{(V_U - V_L)\max}{Vm} \times 100 \qquad (3)$$

Where:
V_L = lower voltage measured in the oscilloscope.
V_U = upper voltage measured in the oscilloscope.
Vm = mean value of the differences between V_U and V_L.
$\Delta V_{max} = (V_U - V_L) \max$

5.- ADMINISTRATION OF THE TEST REPORTS

5.1 Numbering of the reports.

All the generated technical test reports must have a unique and consecutive number.

5.2.- Personnel.

The test of detectors and electronic modules must be done by trained personnel.

5.3.- Technical Report of results.

5.3.1 The results of the test of detectors and electronic modules must be registered in a unique test report, stating the description of the detectors or/and electronic modules, mark, model, serial number, and all the test conditions, including the name of the person who made the tests.

5.3.2 All the test reports must be classified and keep in a folder for future consult.

6.- ACTION IN CASE OF NON CONFORMITIES.

6.1 Technical Report.

Even in the case that results of the test are not as expected, a technical report has to be elaborated, indicating the non conformities and how far are the measured characteristics from the ideal ones.

6.2 Labelling.

The components or equipments that are not under specifications or with a failure have to be marked with a label indicating: **OUT OF SPECIFICATIONS** and **FAILURE** respectively.

7.- RESPONSIBILITIES

7.1.- Head of the Department.

Supervise that all the activities for testing of detectors and electronic modules follow the established procedure.

7.2.- Area Responsible.

7.2.1 Assure that all the electronic test equipment be in good operational conditions and calibrated.

7.2.2 Verify that all the activities for testing of detectors and electronic modules follow the established procedure

7.2.3 Verify that the test reports contain all the details of the testing of detectors and electronic modules.

7.2.4 Maintain a register and control of the test reports for all the detectors and electronic modules tested in the laboratory.

7.3 Operative Personnel.

7.3.1 Verify that all the electronic test equipment be in good operational conditions and calibrated.

7.3.2 Follow the steps established in this procedure for the testing of detectors and electronic modules.

7.3.3 Elaborate the test report of all the tests of the detectors and electronic modules.

7.3.4 Inform to the Area Responsible of any anomalous condition encountered during the test procedure.

8.- BIBLIOGRAPHY

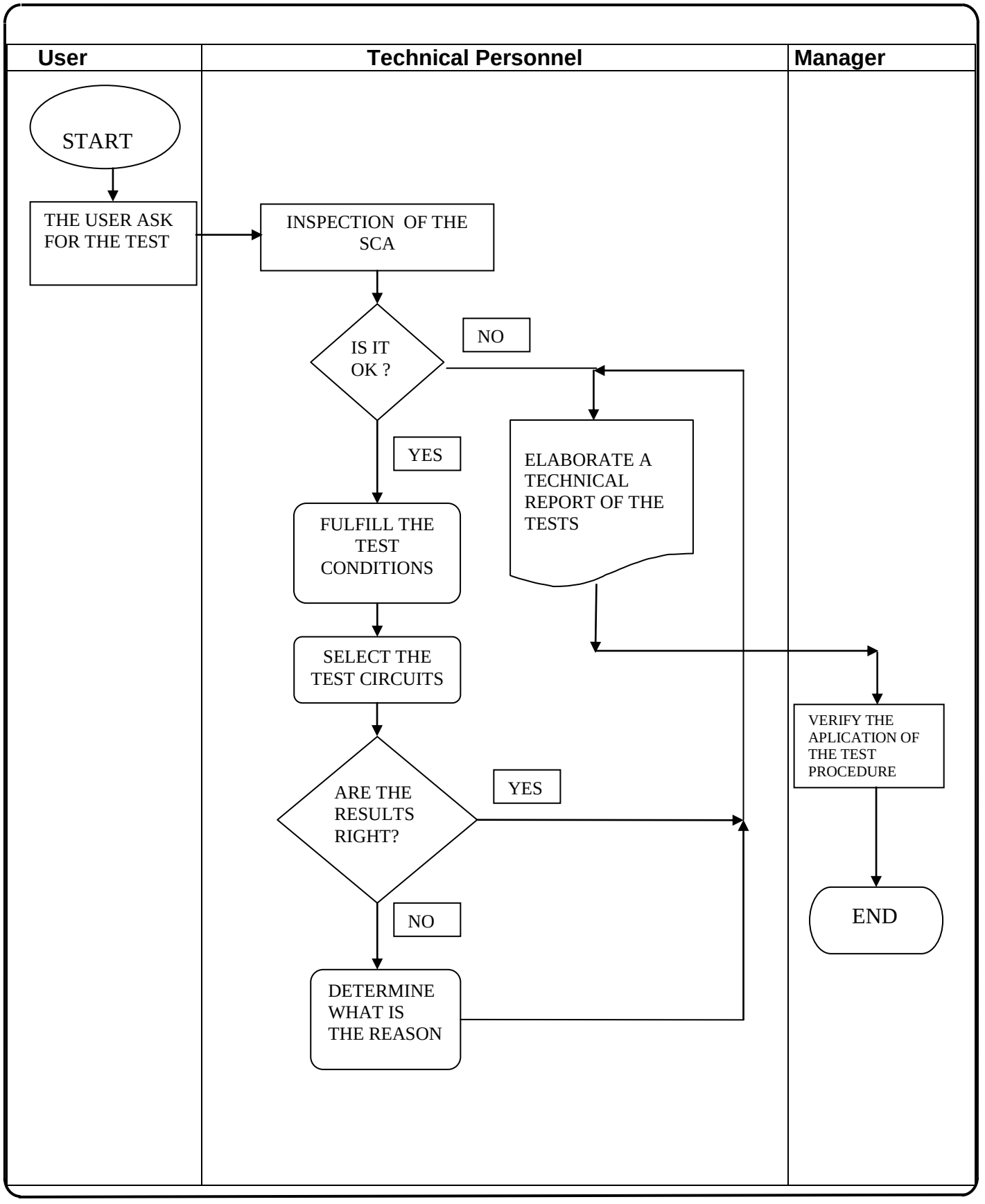
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9.- ANNEXES

Annex I . Flow Chart
Annex II. Test Report

Annex I . Flow Chart


User
Technical Personnel
Manager
START
THE USER ASK FOR THE TEST
INSPECTION OF THE SCA
IS IT OK ?
NO
YES
ELABORATE A TECHNICAL REPORT OF THE TESTS
FULFILL THE TEST CONDITIONS
SELECT THE TEST CIRCUITS
ARE THE RESULTS RIGHT?
YES
NO
DETERMINE WHAT IS THE REASON
VERIFY THE APLICATION OF THE TEST PROCEDURE
END

Annex II. Test Report

TEST REPORT Nº SCA_____

Mark:_____ Model:_____
Serial Number:_____

Instruments Employed

Instrument	Mark	Model	Serial number
Amplifier			
Oscilloscope			
Pulse generator			

Environmental Conditions

Maximum change in temperature	Temperature

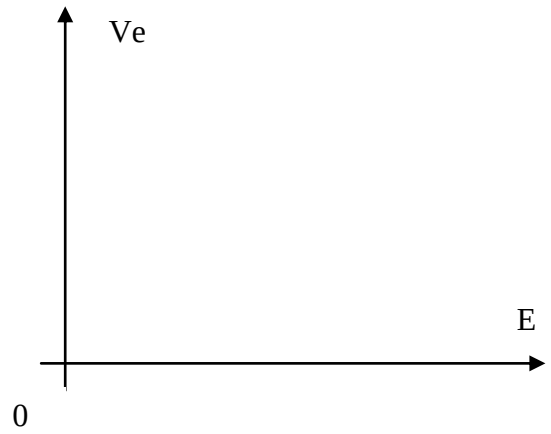
Test Circuit

Fig. 3	

Integral non linearity, INL of the dial E:

E setting	Ve (V) measured	Vi (V) calculated	ΔV (V) calculated
0.5			
1.0			
2.0			
3.0			
4.0			
5.0			
6.0			
7.0			
8.0			
9.0			
10.0			

Table 1. Values to calculate the INL of the E Dial.



D	V_L (V)	V_U (V)	$\Delta V = V_U - V_L$ (V)
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
			Vm =

Table 2. Values to calculate the precision of the window ΔE.

Output voltage (V)	
Pulse width (ns)	
INL of E (%)	
Precision of ΔE (%)	

Diagnostic or Comments:

Tested By:	
Date:	